



Master Module Virtual Product Development

Project Phase 3

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Under Guidance:

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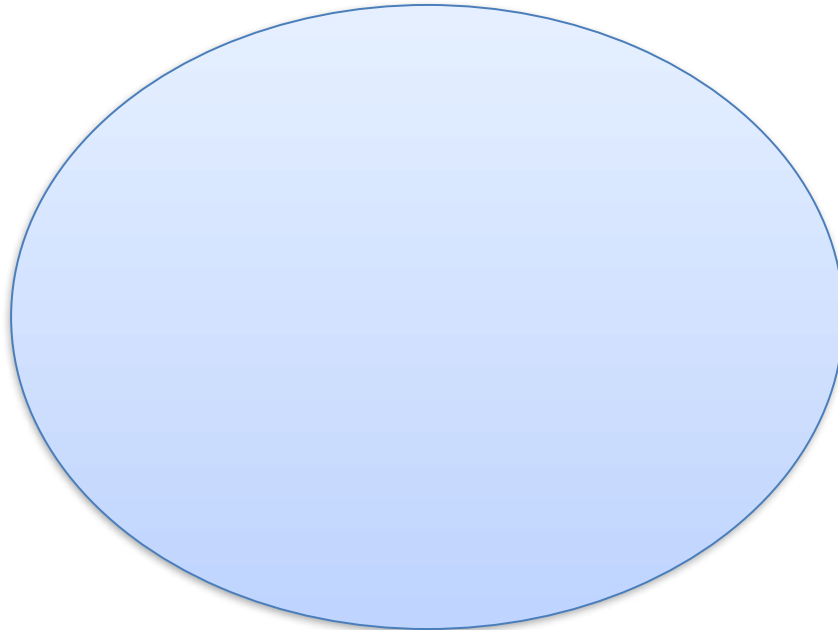
Dipl. Ing. (FH) Arina Engels

1. Final CAD Design
2. Topology Optimization and Stress Analysis
3. Multibody Simulation
4. Actuator Proof
5. Outlook

1. FINAL CAD MODEL



Initially there was...

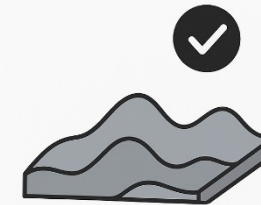


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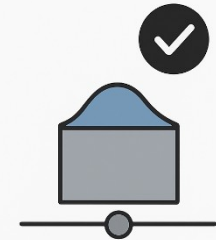


Starting Point : Phase 1

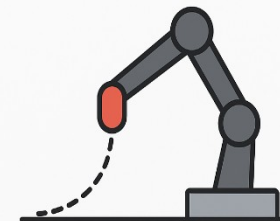
- Requirements:** Define based on core points and project assignment.
- Core Points:** Identify key aspects as the initial step.



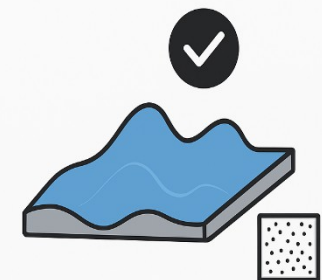
SURFACE FINISH



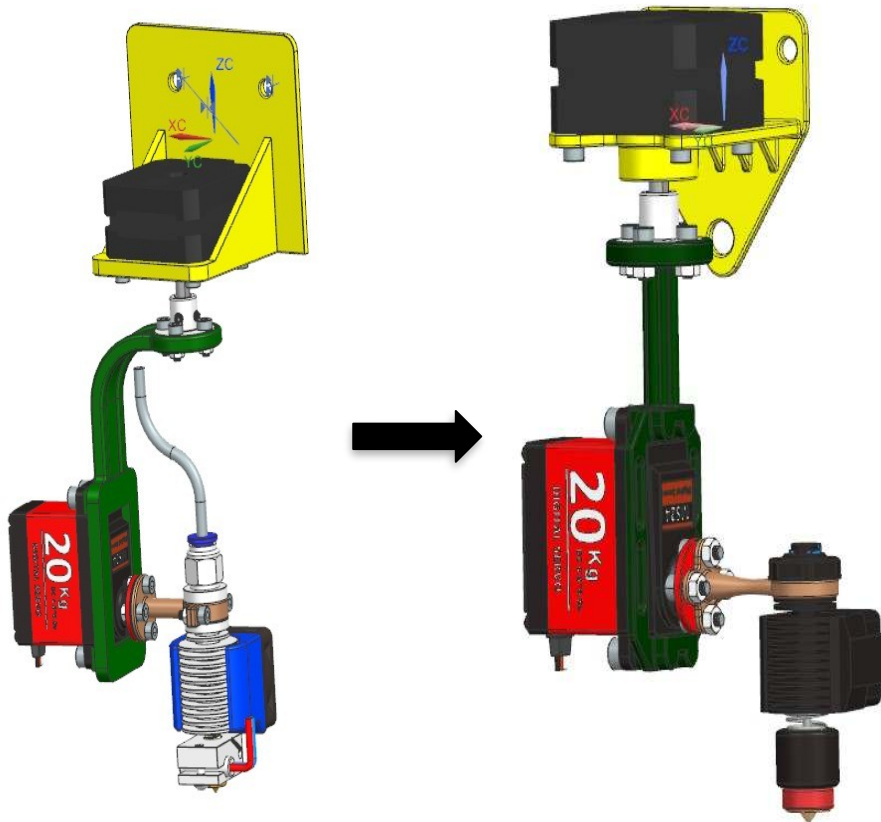
STABILITY



REACHABILITY



SURFACE QUALITY



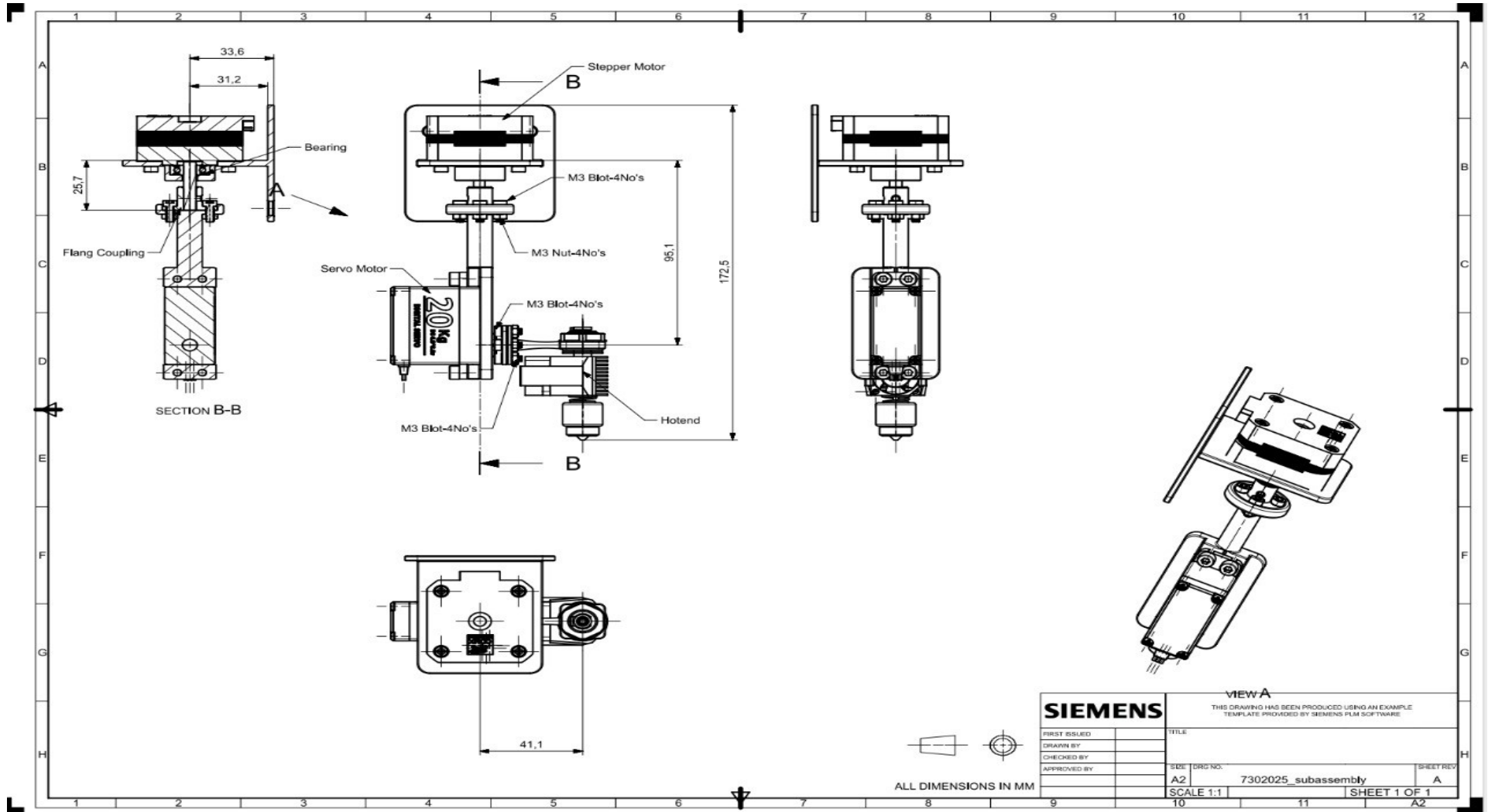
Initial
Model

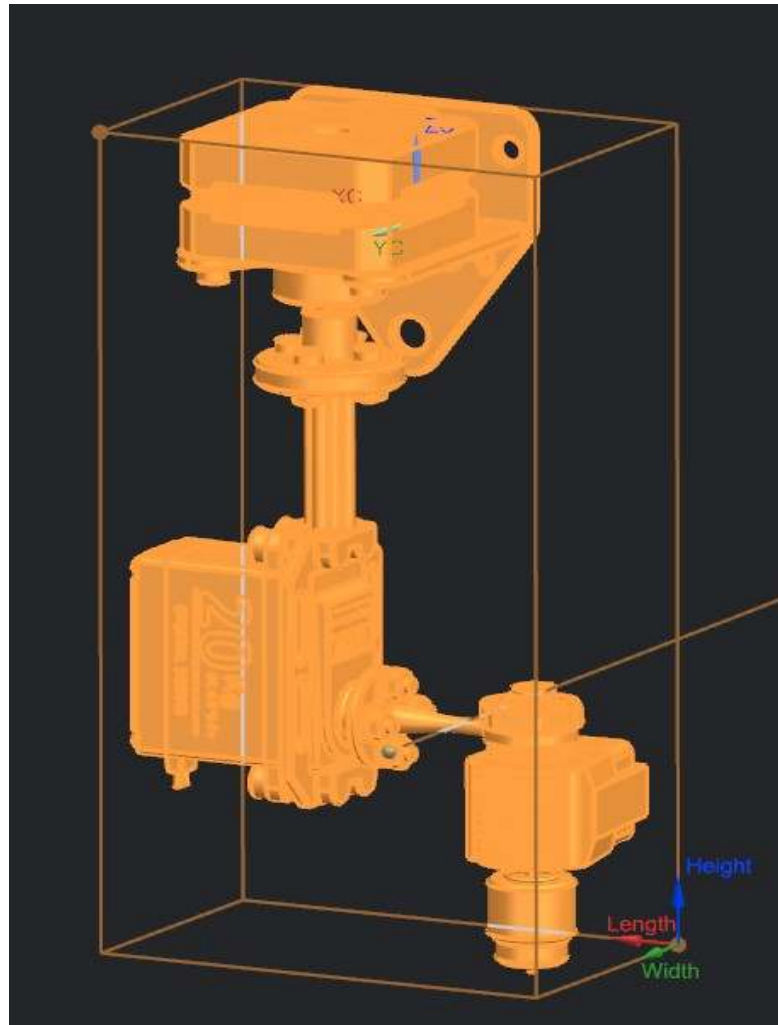
Final
Model

Part Name	Weight
Support Plate	0.0283 Kg
Nema 17 Stepper Motor	0.1320 Kg
Main Frame	0.0179 Kg
Servo Motor	0.0600 Kg
Mg996 r Round Servo Horn	0.0016 Kg
Hotend Mount	0.0029 Kg
E3D Revo Hotend	0.0955 Kg

Overall Weight of Final Printer 0.3382 Kg.

Dimensions





- Centre of Mass : (-7.4935,28.7115,-40.5810) mm
- Length : 88.9045 mm
Width : 54.8500 mm
Height : 168.1043 mm

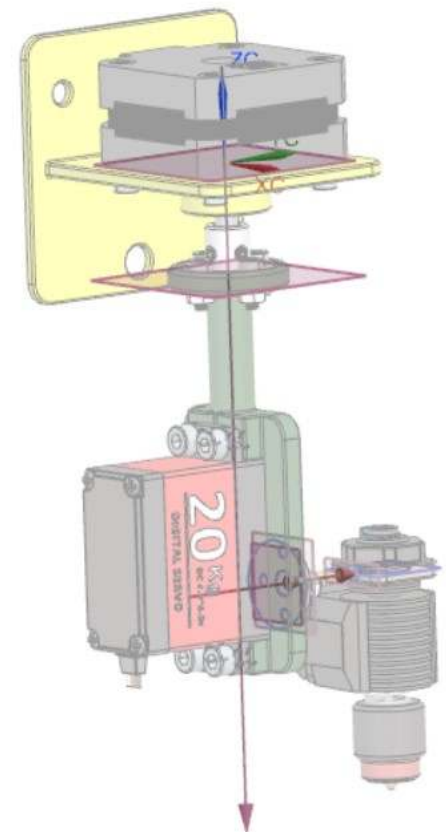
Final CAD Design



Assembly Structure



Optimized Assembly



Skeleton Structure

Changes Done in Third Phase

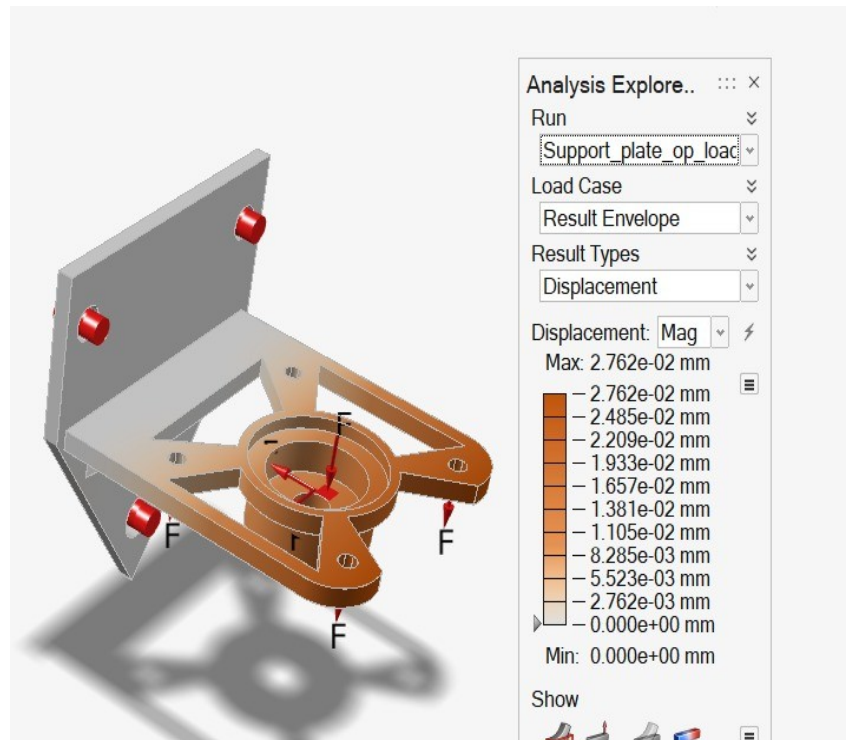


After Phase 2 reviews, we decided to update the hot end. We selected the E3D Revo Micro, which is more compact than the previous design, with considerations for weight and cost.

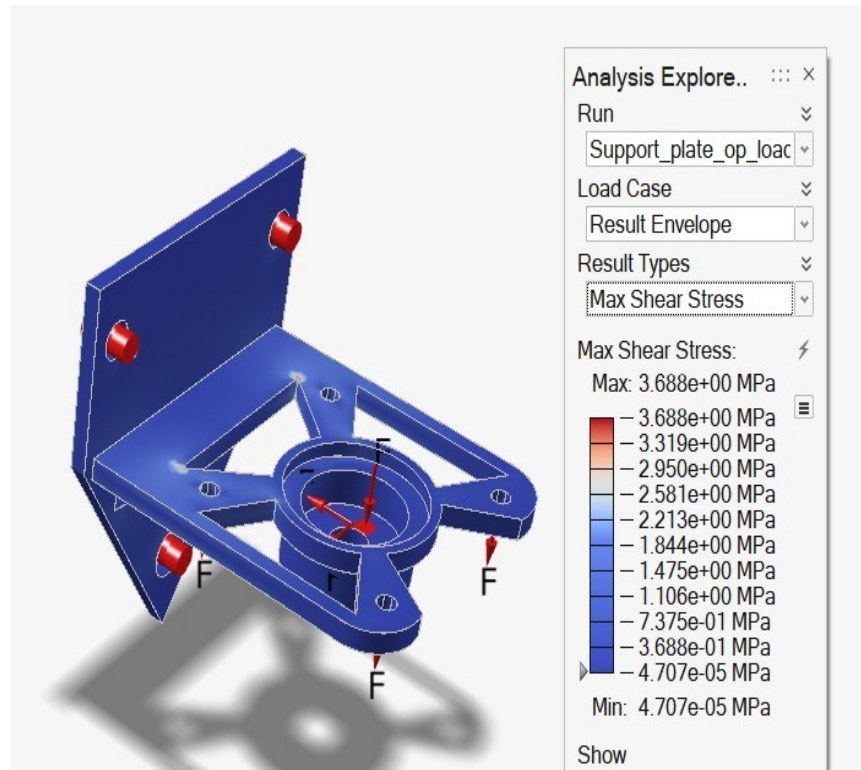
2. Topology Optimization

Due to hot end changes and minor CAD modifications, we performed topology optimization for the support plate, mainframe, and hot end mount, while imported parts like the rigid flange coupling and MG996R servo horn were excluded.

1. Support Plate



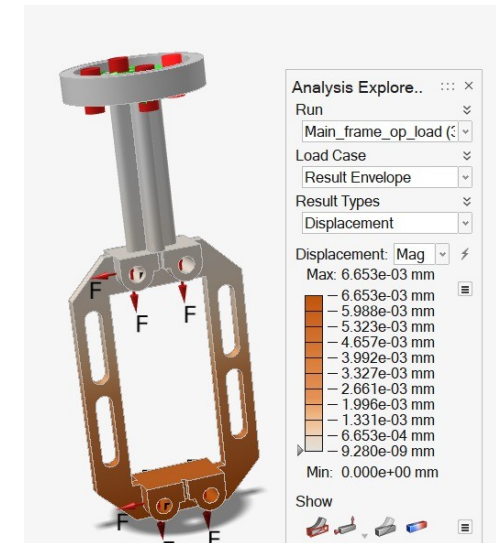
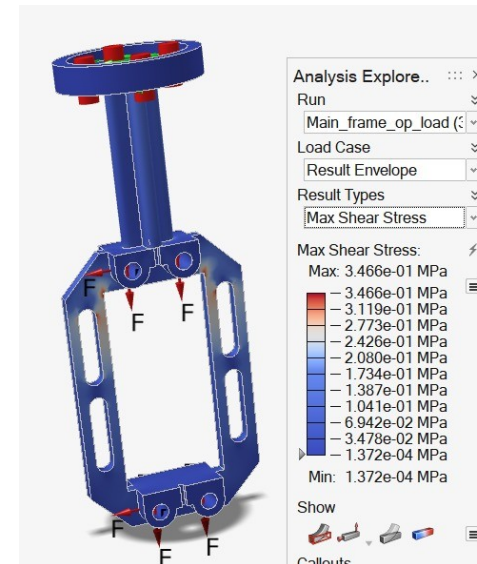
Description	Initially	Final
Mass	0.0459 Kg	0.0283 Kg
Deformation	0.01683 mm	0.0276 mm



Description	Initially	Final
Max Shear Stress	1.226 MPa	3.688 MPa

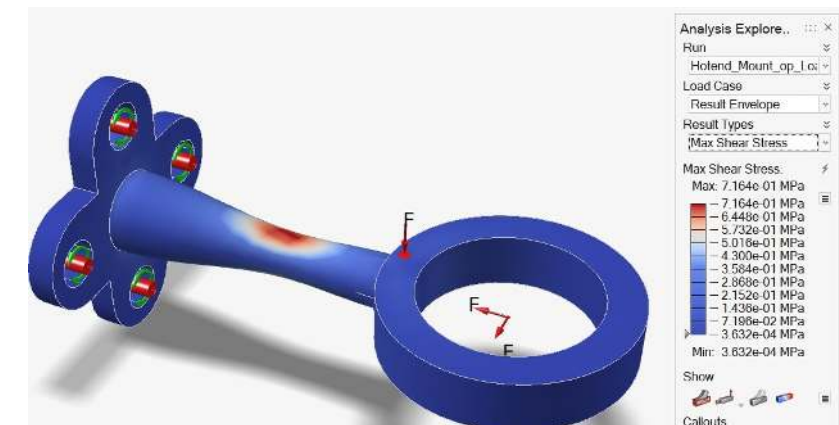
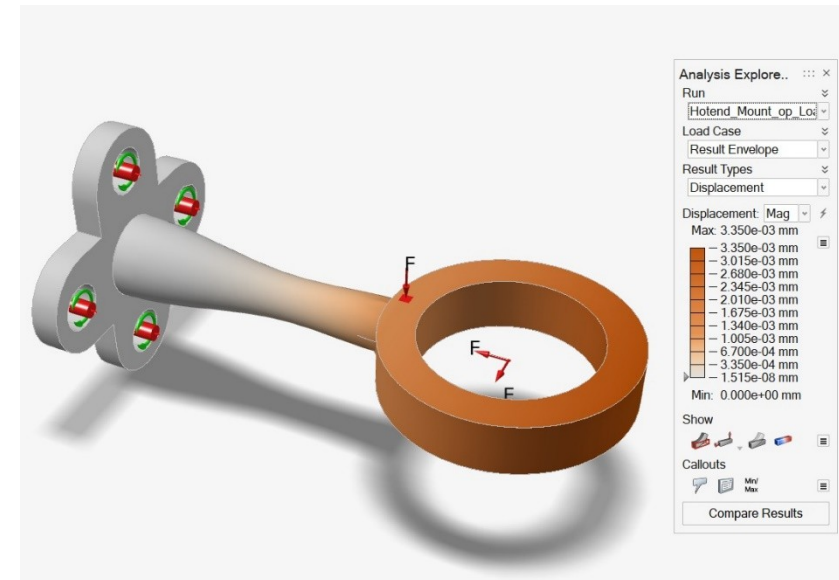
2. Main Frame

Description	Initially	Final
Mass	0.0305 Kg	0.0179 Kg
Max.Shear Stress	0.73Mpa	0.346 MPa
Deformation	0.005mm	0.0066mm

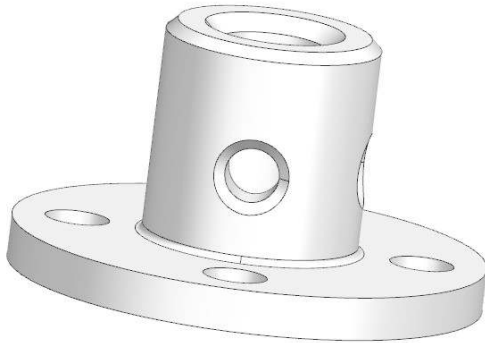


3. Hotend Mount

Description	Initially	Final
Mass	0.0043Kg	0.0029Kg
Max.Shear Stress	0.75Mpa	0.71MPa
Deformation	0.0039mm	0.0033mm



4. Rigid Flange Coupling



5. Mg996r round horn for servo



The rigid flange coupling (0.033 kg) and MG996R horn (0.0016 kg) are imported, lightweight parts that serve the model as is and do not require optimization.

Radial and Axial Play

The NEMA 17 main shaft can handle up to 450 g of radial or axial load. Our model applies only 199 g, which is well below the limit. Therefore, the shaft operates safely within its capacity.

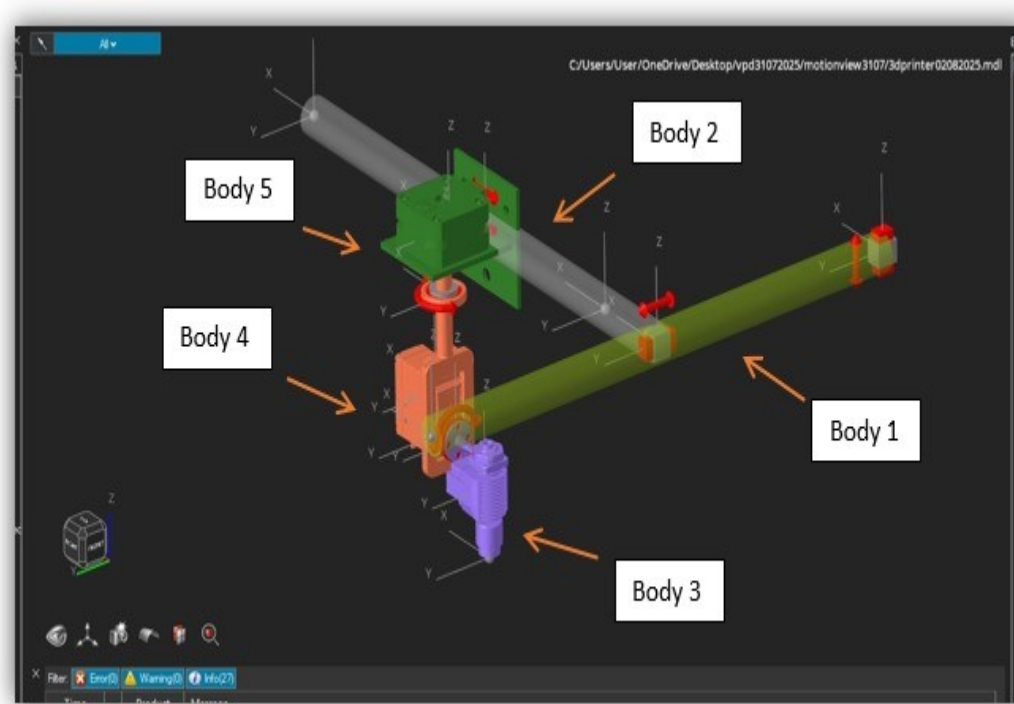
Insulation Class		B										
Dielectric Strength		500VAC for one minute										
Shaft Radial Play		0.06Max. (450 g-load)										
Shaft Axial Play		0.08Max. (450 g-load)										

Technique Specification												
Model		Step angle	Motor length	Rated voltage	Rated current	Phase resistance	Phase inductance	Holding torque		lead wire	Rotor inertia	Weight
Single Shaft	Double Shaft	*	L(mm)	V	A	Ω	mH	Oz-in	gf.cm	No.	g.cm ²	kg
42BYG03-021A	42BYG03-001B	1.8	33	12	0.3	40	22	23.5	1700	5	35	0.22
42BYG03-021A	42BYG03-001B	1.8	33	4	0.95	4.2	2.5	22	1580	6	35	0.22
42BYG03-022A	42BYG03-002B	1.8	33	9.6	0.4	24	15	22	1580	6	35	0.22
42BYG03-023A	42BYG03-003B	1.8	33	12	0.31	38.5	21	22	1580	6	35	0.22
42BYG03-021A	42BYG03-001B	1.8	33	2.8	1.33	2.1	2.5	30	2200	4	35	0.22
42BYG03-021A	42BYG03-001B	1.8	39	4	1.2	3.3	3.2	36	2590	6	54	0.28
42BYG03-022A	42BYG03-002B	1.8	39	6	0.8	7.5	6.7	36	2590	6	54	0.28
42BYG03-023A	42BYG03-003B	1.8	39	12	0.4	30	30	36	2590	6	54	0.28
42BYG03-021A	42BYG03-001B	1.8	39	2.8	1.68	1.65	3.2	50	3600	4	54	0.28
42BYG03-021A	42BYG03-001B	1.8	47	4	1.2	3.3	2.8	44	3170	6	68	0.35

https://tse1.mm.bing.net/th/id/OIP.Fj0rJCN9Ipr6_0zy_IYWHwHaI8?pid=Api&P=0&h=180

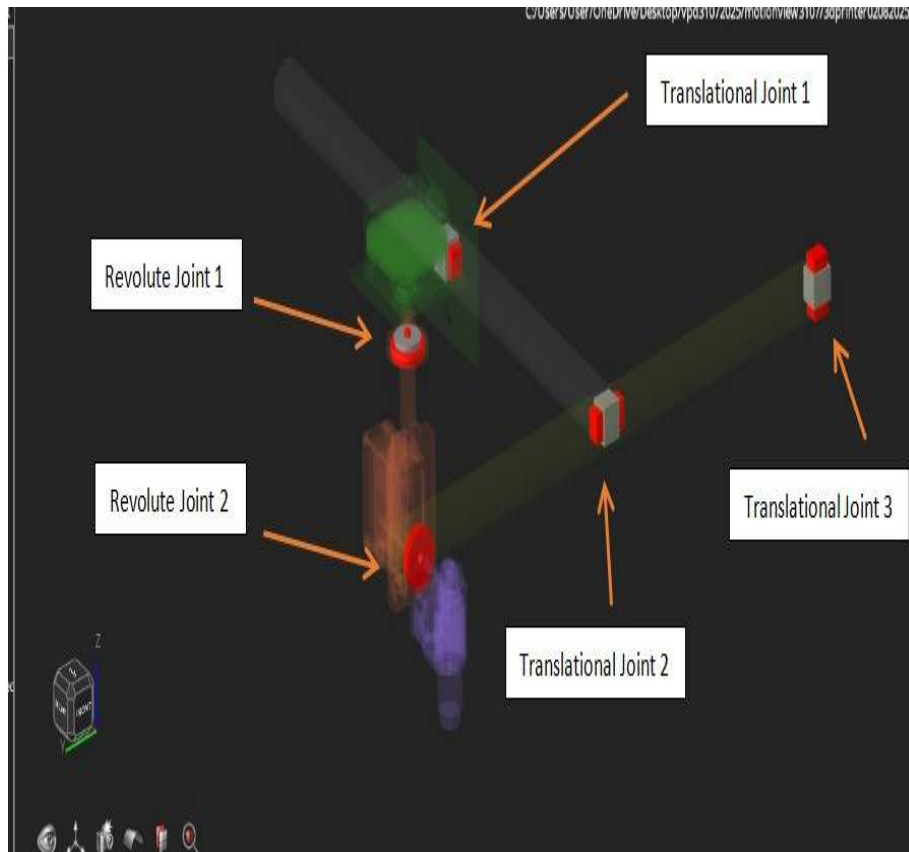
3. MULTI BODY SIMULATION

Imported CAD Model



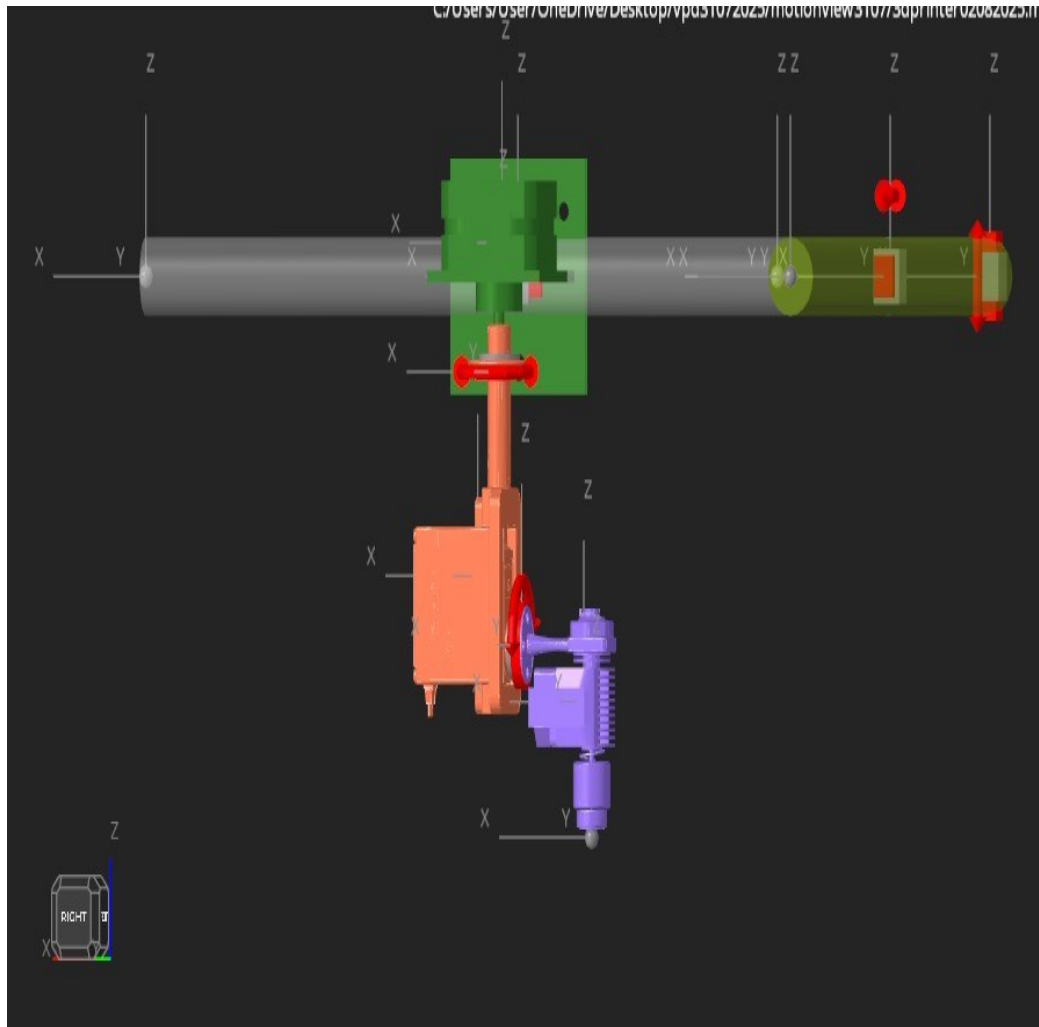
- Body 1: Y–rail
- Body 2: X–rail
- Body 3: Hot end and hot end holder
- Body 4: Servo motor and main frame
- Body 5: Support plate and NEMA motor

Joints



- Revolute Joint 1: Rotates finger printer around global Z-axis
- Revolute Joint 2: Rotates hot end mount around global Y-axis
- Translational Joint 1: Moves finger printer along global X-axis
- Translational Joint 2: Moves finger printer along global Y-axis
- Translational Joint 3: Moves bed along global Z-axis

Arrangement of CAD Model



Markers (13)		
Global Frame	Global_Frame	30101010
mk_yrail_center0	mk_yrail_center0	30102021
mk_yrail_end	mk_yrail_end	30102030
mk_yrail_start	mk_yrail_start	30102040
mk_xrail_center	mk_xrail_center	30103070
mk_xrail_end	mk_xrail_end	30103060
mk_cm_supportnema	mk_cm_supportnema	30106090
mk_cm_mainservo	mk_cm_mainservo	30105100
mk_cm_hotend	mk_cm_hotend	30104110
mk_endeffector	mk_endeffector	30104140
mk_rev_nema_mainframe	mk_rev_nema_mainframe	30106121
mk_rev_mainframe_hotend	mk_rev_mainframe_hotend	30105131
Marker 0	m_0	30103180
Outputs (13)		

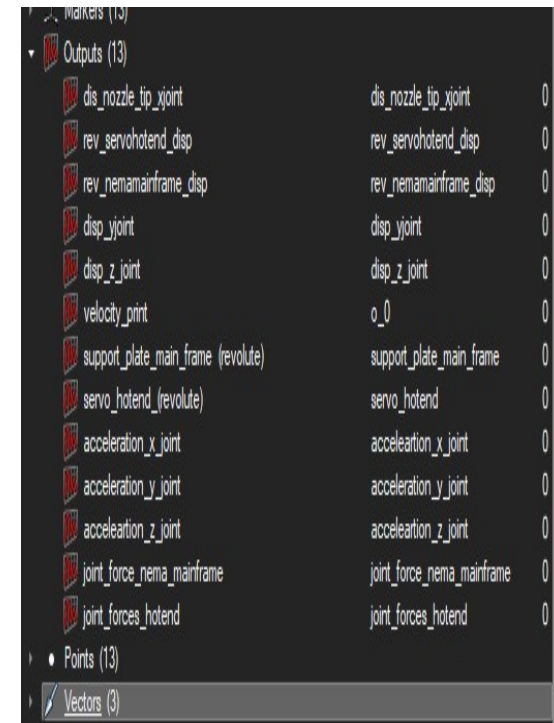
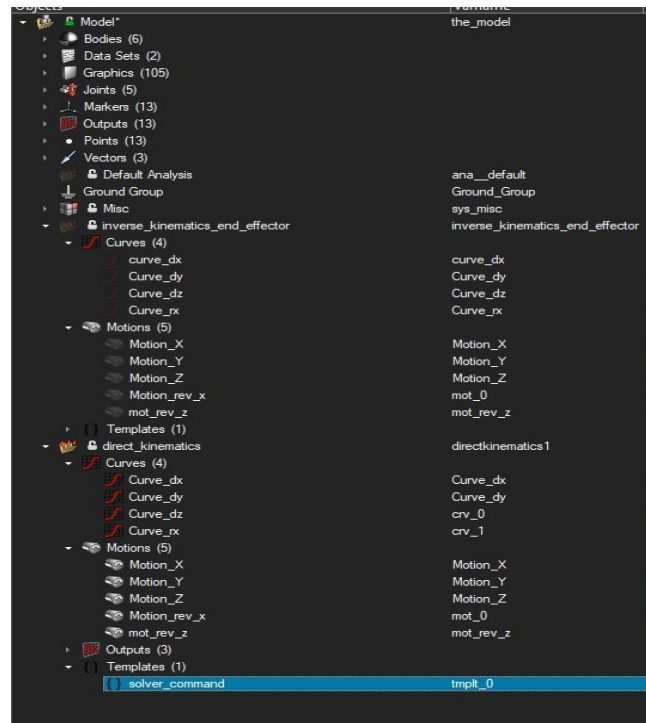
Markers

joint_force_nema_mainframe	joint_force_nema_mainframe	70000042
joint_forces_hotend	joint_forces_hotend	70000044
Points (13)		
Global Origin	P_Global_Origin	301001
pt_y_center0	pt_y_center0	301002
pt_y_end	pt_y_end	301003
pt_y_start	pt_y_start	301004
pt_xrail_end0	pt_xrail_end0	301006
pt_xrail_center0	pt_xrail_center0	301007
pt_cm_support_nema0	pt_cm_support_nema0	301009
pt_cm_mainservo0	pt_cm_mainservo0	301010
pt_cm_hotend	pt_cm_hotend	301011
pt_rev_nemamainframe	pt_rev_nemamainframe	301012
pt_rev_servohotend	pt_rev_servohotend	301013
pt_end_effector	pt_end_effector	301014
p_shifted_origin0	p_shifted_origin0	301018
Vectors (3)		
Default Analysis	ana_default	
Ground Group	Ground_Group	

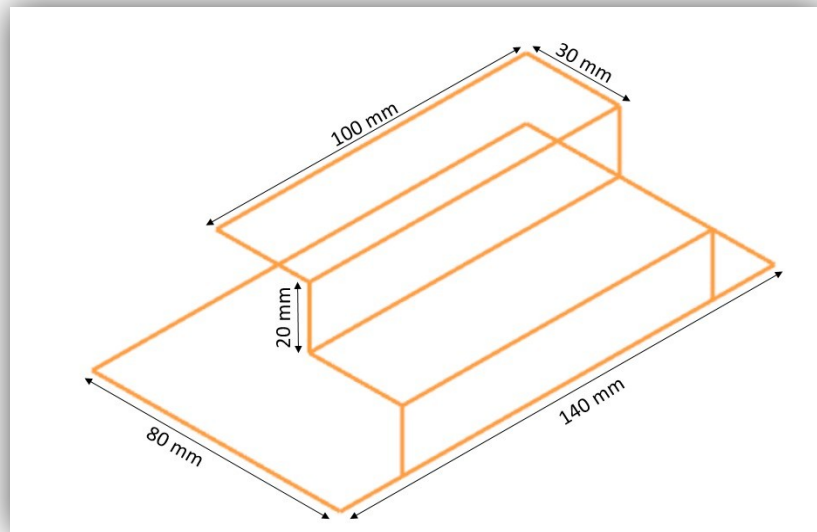
Points

Input and Outputs for Staircase Example

	A	B	C	D	E	F
1	time	x	y	z	rx	
2	0	0.05	0	0	0	
3	0.001563	0.05	-0.00081	0	0	
4	0.003125	0.05	-0.00162	0	0	
5	0.004688	0.05	-0.00242	0	0	
6	0.00625	0.05	-0.00323	0	0	
7	0.007813	0.05	-0.00404	0	0	
8	0.009375	0.05	-0.00485	0	0	
9	0.010938	0.05	-0.00566	0	0	
10	0.0125	0.05	-0.00646	0	0	
11	0.014063	0.05	-0.00727	0	0	
12	0.015625	0.05	-0.00808	0	0	
13	0.017188	0.05	-0.00889	0	0	
14	0.01875	0.05	-0.0097	0	0	
15	0.020313	0.05	-0.01051	0	0	
16	0.021876	0.05	-0.01131	0	0	
17	0.023438	0.05	-0.01212	0	0	
18	0.025001	0.05	-0.01293	0	0	
19	0.026563	0.05	-0.01374	0	0	
20	0.028126	0.05	-0.01455	0	0	
21	0.029688	0.05	-0.01535	0	0	
22	0.031251	0.05	-0.01616	0	0	
23	0.032813	0.05	-0.01697	0	0	
24	0.034376	0.05	-0.01778	0	0	
25	0.035938	0.05	-0.01859	0	0	
26	0.037501	0.05	-0.01939	0	0	
27	0.039064	0.05	-0.0202	0	0	
28	0.040626	0.05	-0.02101	0	0	
29	0.042189	0.05	-0.02182	0	0	
30	0.043751	0.05	-0.02263	0	0	

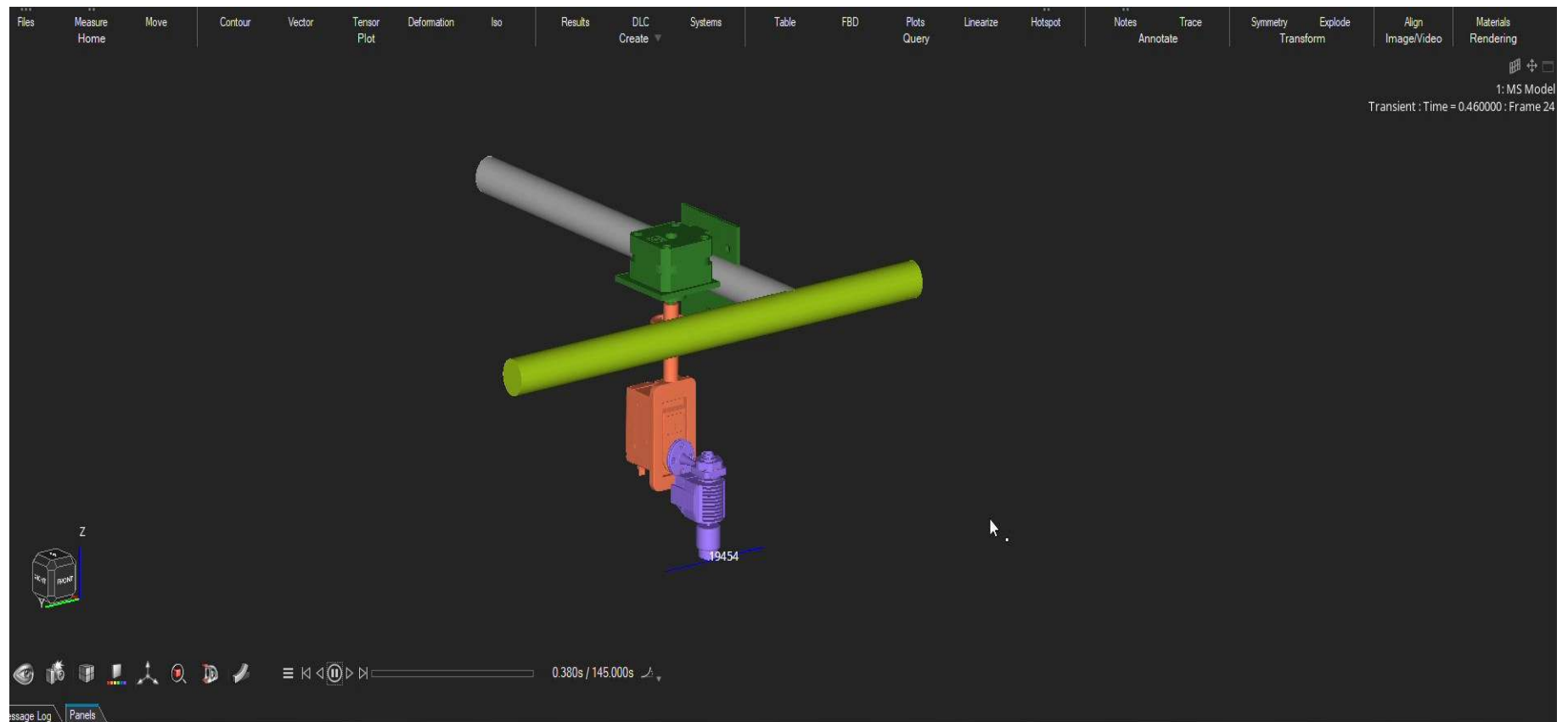


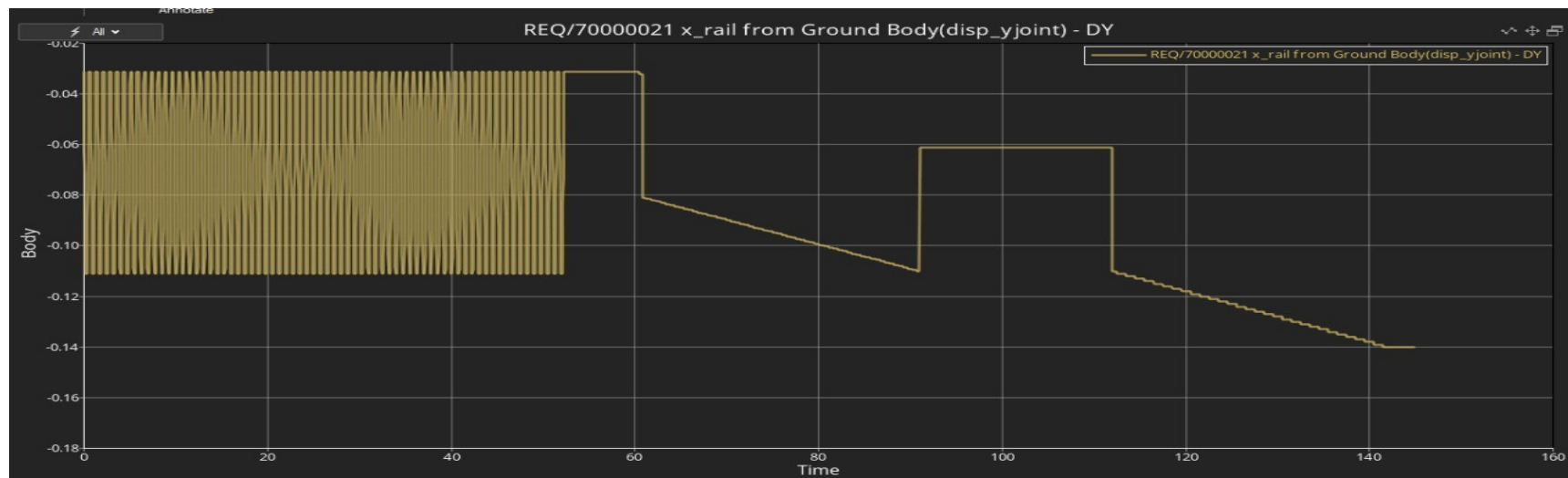
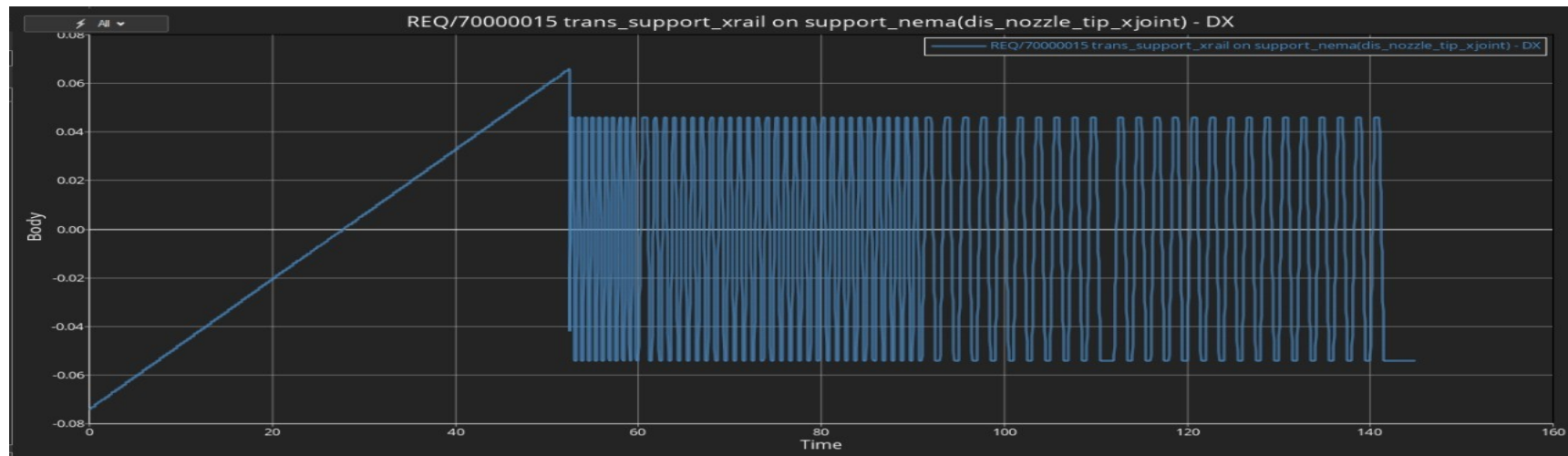
- Total Degrees of Freedom from 5 bodies = 30
- Degrees of Freedom constrained by the joints = 25
- Resultant degrees of freedom of the nozzle = 5 (Translational along X, Y, Z and Rotational about X and Z axes).



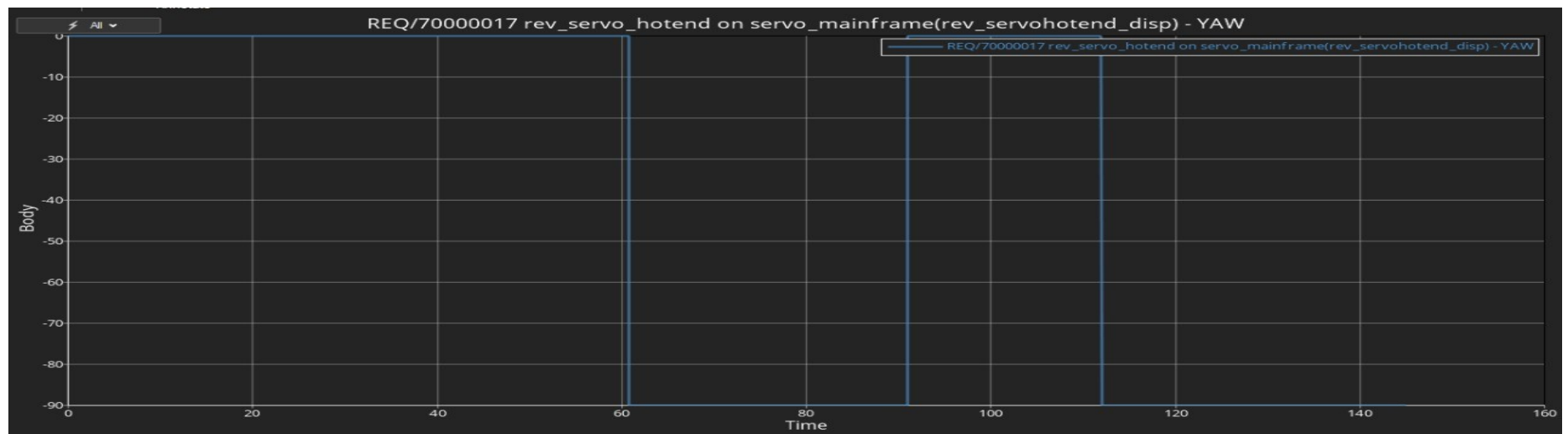
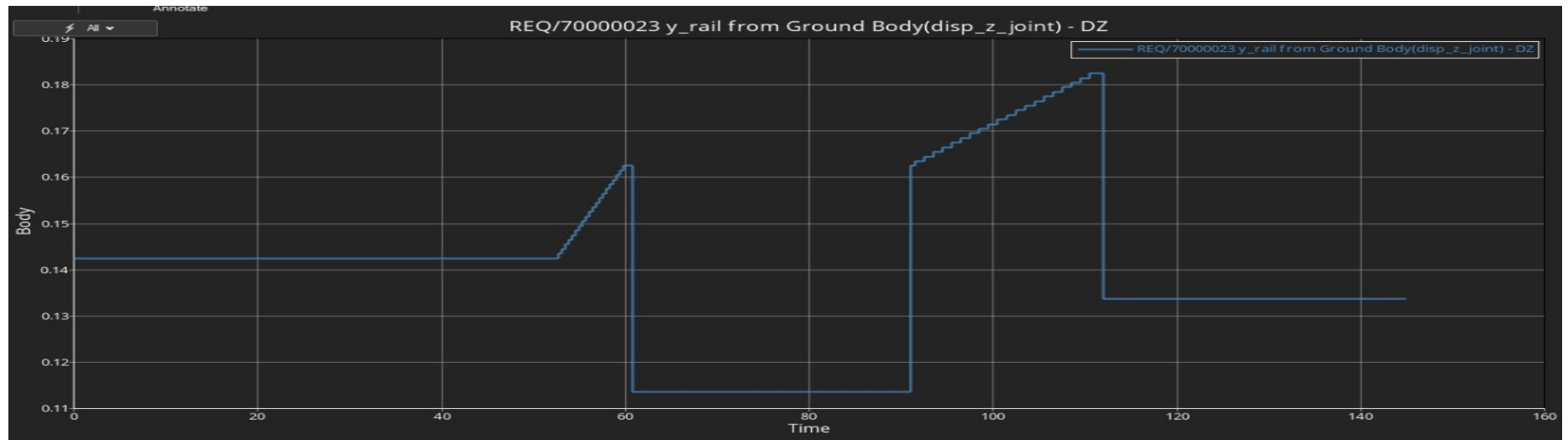
- The staircase was printed with smooth curve-based motion, with gradual nozzle tilting in the second phase for precise, seamless layers.

Results and Graphs



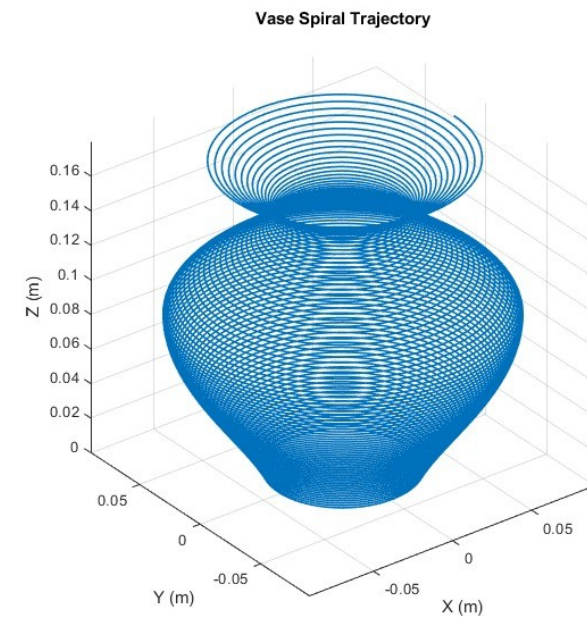
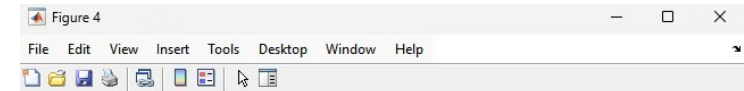
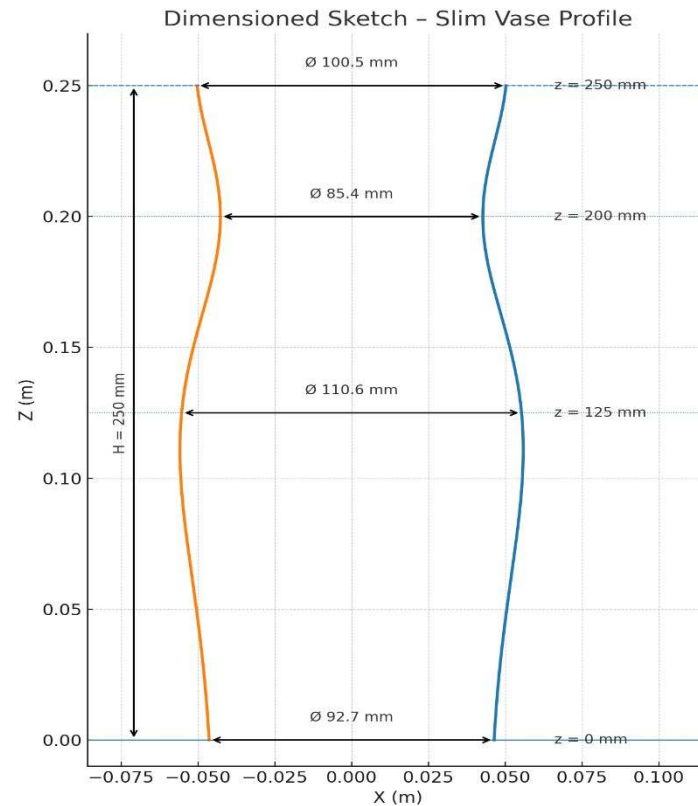


Displacement X and Y



Displacement Z and Rotational X

Self Defined Example – Flower Vase



Expressions for Trajectory

Translational Motion in X - $0.165 + (0.045 + (0.04025) * \exp(-((0.002777778 * \text{TIME} - 0.125)^2 / (2 * 0.06^2)) - (0.02145) * \exp(-((0.002777778 * \text{TIME} - 0.200)^2 / (2 * 0.04^2)) + (0.0182) * \exp(-((0.002777778 * \text{TIME} - 0.250)^2 / (2 * 0.03^2))) * \cos(43.6332313 * \text{TIME})$

Translational Motion in Y - $(0.045 + (0.04025) * \exp(-((0.002777778 * \text{TIME} - 0.125)^2 / (2 * 0.06^2)) - (0.02145) * \exp(-((0.002777778 * \text{TIME} - 0.200)^2 / (2 * 0.04^2)) + (0.0182) * \exp(-((0.002777778 * \text{TIME} - 0.250)^2 / (2 * 0.03^2))) * \sin(43.6332313 * \text{TIME})$

Expressions for Trajectory

Translational Motion in Z - $0.002777778 * \text{TIME}$

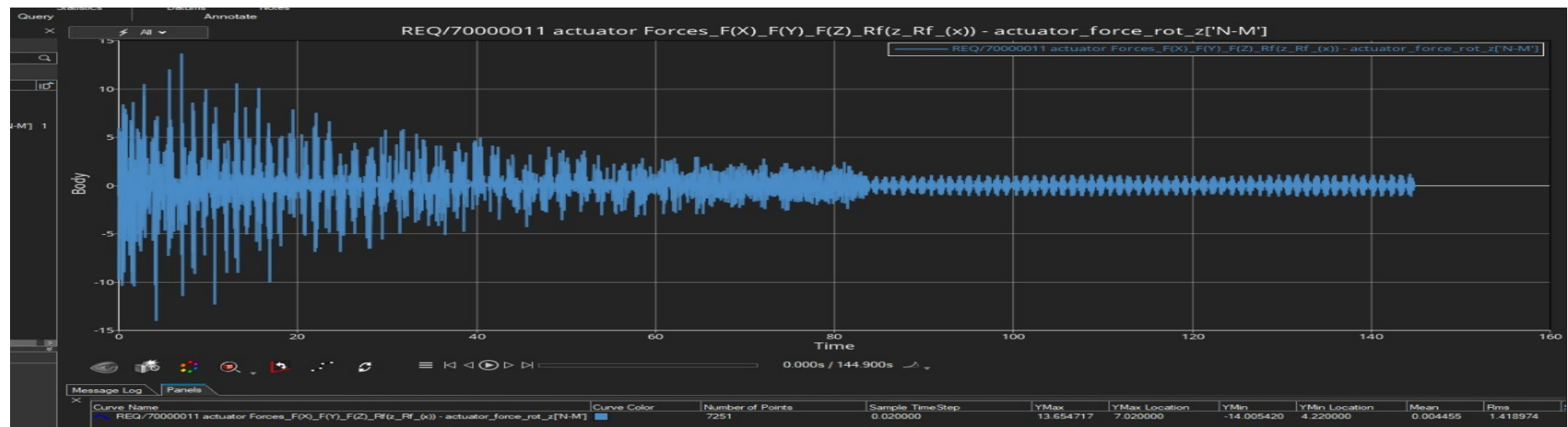
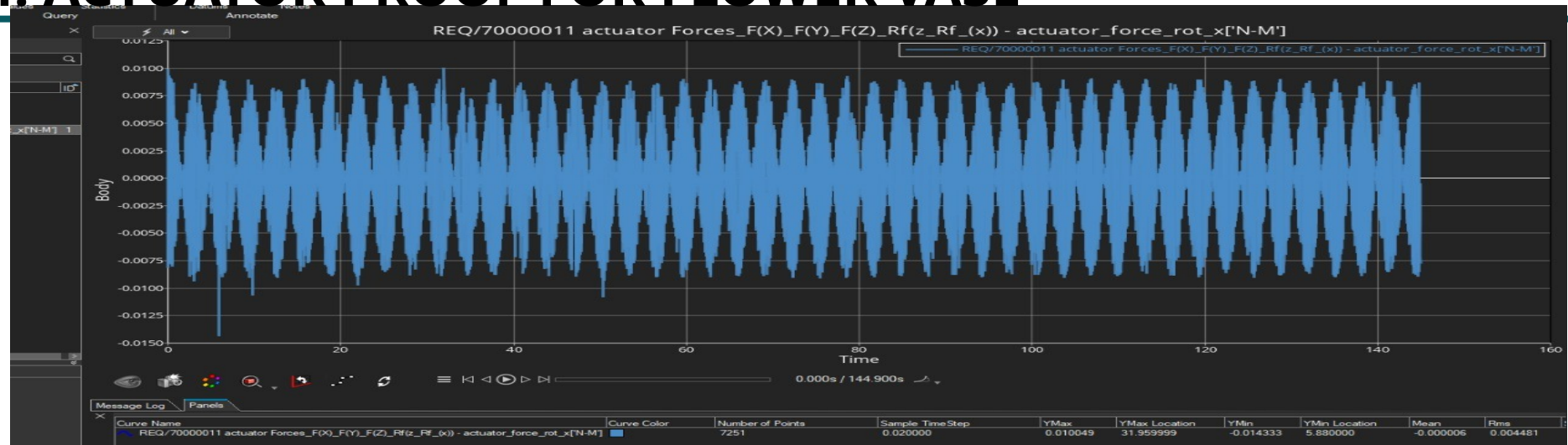
Rotational Motion in Z - $(43.6332313 * \text{TIME}) + 1.570796327$

Rotational Motion in X - 0.785398163

Results and Graphs

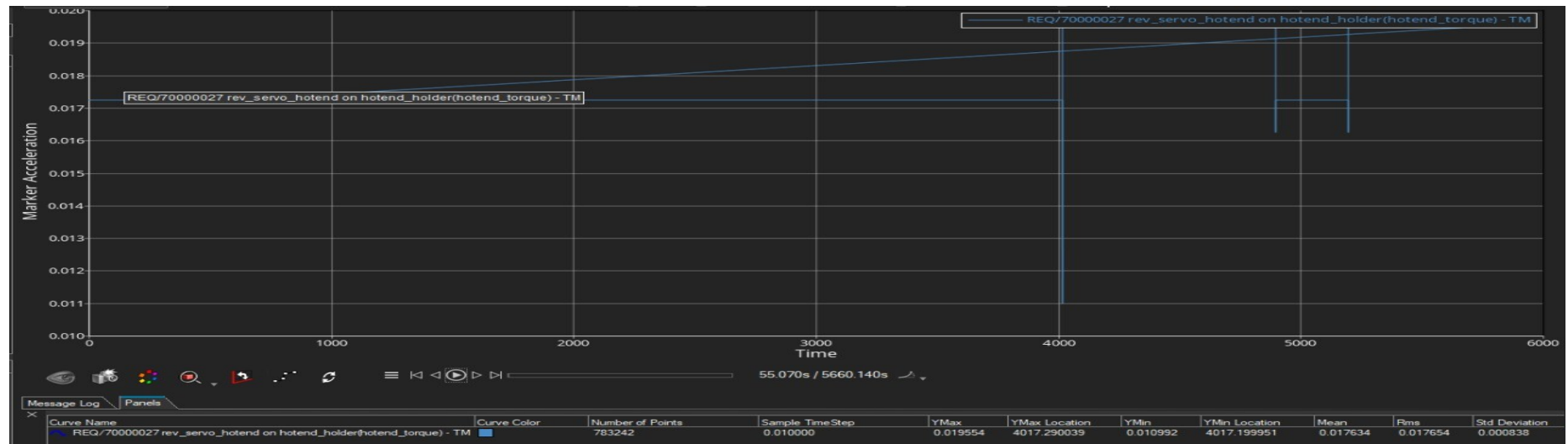
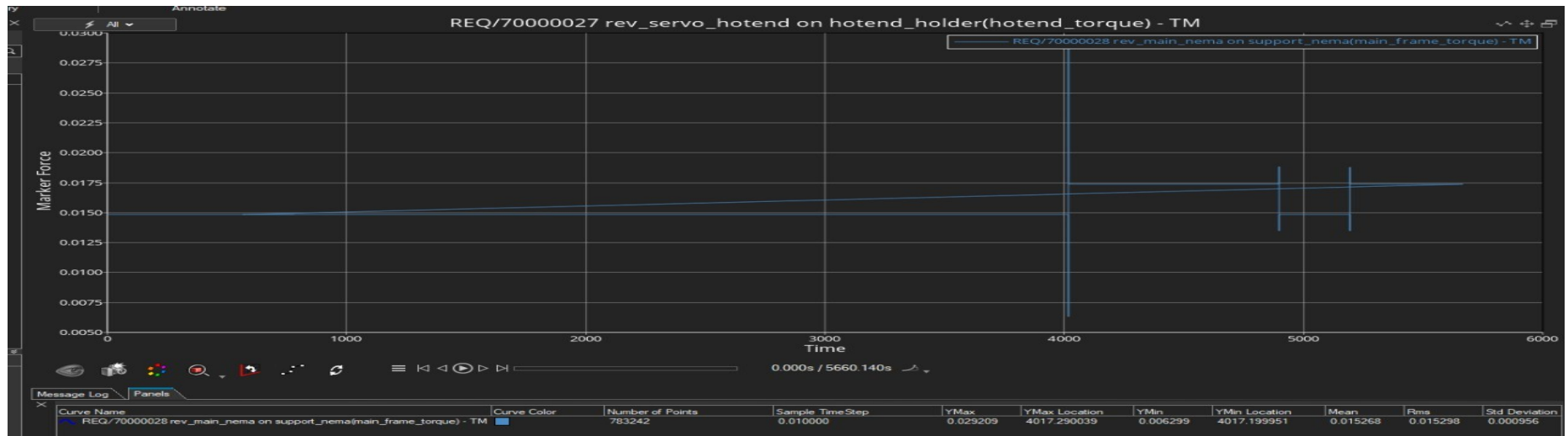


4. ACTUATOR PROOF FOR FLOWER VASE



Actuator forces in rotational X and Z

ACTUATOR PROOF FOR STAIRCASE



Actuator Forces for Nema and Servo Motor

5. OUTLOOK



- Fastener unification for cost efficiency.
- Light weighting via topology optimization.
- Design for manufacturability (DFM).
- Motion tuning with torque profiles.
- Multi-material / filament strategies.
- Integration with 6-axis robotics

Lessons Learned From Project

Besides the software skills in Siemens NX, Altair Inspire and Motionview, we learned:

- Project planning and time management.
- Technical and English communication.
- Structured problem-solving and decision-making.
- Presenting results to stakeholders.
- Team collaboration and coordination.
- Adaptability and working under deadlines



THANK YOU ANY QUESTIONS!!